

Model dossier — CHEIDI

Status: cross-check missing. The blank-slate reconstruction completed and is included below, but the paper cross-check never ran, so no term-by-term comparison exists.

This dossier is not a completed reverse-check review package. It collects what exists for this model so you can read the physics without opening the source files. Where a full review package does exist for a model, it is the `REVIEW.pdf` in that model's directory.

item	value
model	CHEIDI
chain status	passed the full chain
Lagrangian source	repair2/final.fr (re-validated)
Lagrangian terms found	12

Verbatim Lagrangian terms

Quoted unmodified from the `.fr`. These are the terms any reconstruction would have to account for.

`heidiVDefault (=)`

`246.22`

`heidiMhDefault (=)`

`120.`

`heidiCsDefault (=)`

`100.`

`heidiMbDefault (=)`

`100.`

`heidiG2Default (=)`

`1.`

`heidiCutoffDefault (=)`

`1000.`

`heidiNmodesDefault (=)`

`10`

`heidiModeRange (=)`

`Range[heidiNmodesDefault]`

`LHEIDI (:=)`

`(LSM /. {muH -> 0, \[Lambda] -> 0}) /. H -> (Hh[n]*xi[n])`

LHEIDIgg (:=)

```
Block[{fun, piece, tt, mm, hh}, tt[m_] := 4*MT^2/m^2; fun[tau_] := tau*(1 + (1 -
  ↳ tau)*If[NumericalValue[tau] > 1, ArcSin[Sqrt[1/tau]]^2, -1/4*(Log[(1 + Sqrt[1 + tau])/(1
  ↳ - Sqrt[1 - tau])] - I*Pi)^2]); mm[n_] := Symbol["mhh" <> ToString[n]]; hh[n_] :=
  ↳ Symbol["Hh" <> ToString[n]]; piece[n_] :=
  ↳ Sqrt[kggh]*gs^2/32/Pi^2/v*fun[tt[mm[n]]]*xi[n]*hh[n]*(del[G[mu, a], nu] - del[G[nu, a],
  ↳ mu])^2; Plus @@ (piece /@ heidiModeRange)]
```

LHEIDIggHeavyTop (:=)

```
Block[{piece, mm, hh}, mm[n_] := Symbol["mhh" <> ToString[n]]; hh[n_] := Symbol["Hh" <>
  ↳ ToString[n]]; piece[n_] := Sqrt[kggh]*gs^2/32/Pi^2/v*(2/3)*xi[n]*hh[n]*(del[G[mu, a],
  ↳ nu] - del[G[nu, a], mu])^2; Plus @@ (piece /@ heidiModeRange)]
```

LTot (:=)

LHEIDIggHeavyTop

Blank-slate reconstruction

Written by an agent that saw only the sanitized `.fr` — no paper, no model name, no history.

Reconstruction of `sanitized.fr`

Lagrangian

Define the real scalar mode tower

$$S(x) \equiv \sum_{k=1}^{10} \xi_k H h_k(x),$$

where `Hh[k]` is implemented as the class members `Hh1` through `Hh10`.

The new fields `Hh_k` have no declared SM gauge quantum numbers. Their covariant derivative is therefore just

$$D_\mu H h_k = \partial_\mu H h_k.$$

For SM fields appearing implicitly through `LSM`, the usual SM covariant derivative is understood:

$$D_\mu = \partial_\mu + i g_s T^a G_\mu^a + i g \frac{\tau^I}{2} W_\mu^I + i g' Y B_\mu,$$

with the relevant representation matrices omitted for gauge-singlet fields.

LHEIDI

The file defines

$$\text{LHEIDI} = \mathcal{L}_{\text{SM}}|_{\mu_H \rightarrow 0, \lambda \rightarrow 0, H \rightarrow S}.$$

The new-field-dependent part is therefore the SM Higgs interaction structure with the physical Higgs field replaced by $S = \sum_k \xi_k H h_k$, while the Higgs potential terms proportional to `muH` and `\[Lambda]` are removed.

Scalar kinetic term:

$$\mathcal{L}_{\text{LHEIDI}} \supset \frac{1}{2} \partial_\mu S \partial^\mu S = \frac{1}{2} \sum_{k,\ell=1}^{10} \xi_k \xi_\ell \partial_\mu H h_k \partial^\mu H h_\ell$$

Gauge-boson mass-interaction terms:

$$\mathcal{L}_{\text{LHEIDI}} \supset \frac{2M_W^2}{v} S W_\mu^+ W^{-\mu}$$

$$\mathcal{L}_{\text{LHEIDI}} \supset \frac{M_W^2}{v^2} S^2 W_\mu^+ W^{-\mu}$$

$$\mathcal{L}_{\text{LHEIDI}} \supset \frac{M_Z^2}{v} S Z_\mu Z^\mu$$

$$\mathcal{L}_{\text{LHEIDI}} \supset \frac{M_Z^2}{2v^2} S^2 Z_\mu Z^\mu$$

Fermion Yukawa terms:

$$\mathcal{L}_{\text{LHEIDI}} \supset - \sum_i \frac{m_{u_i}}{v} S \bar{u}_i u_i$$

$$\mathcal{L}_{\text{LHEIDI}} \supset - \sum_i \frac{m_{d_i}}{v} S \bar{d}_i d_i$$

$$\mathcal{L}_{\text{LHEIDI}} \supset - \sum_i \frac{m_{\ell_i}}{v} S \bar{\ell}_i \ell_i$$

Equivalently, in chiral-projector notation,

$$\bar{f}f = \bar{f}_L f_R + \bar{f}_R f_L = \bar{f} P_R f + \bar{f} P_L f, \quad P_{R,L} = \frac{1 \pm \gamma^5}{2}.$$

The file removes the SM Higgs potential through

$$\mu_H \rightarrow 0, \quad \lambda \rightarrow 0,$$

so no S , S^2 , S^3 , or S^4 potential terms are generated by **LHEIDI**.

LHEIDIgg

The file also defines a finite-top-mass effective scalar-gluon-gluon interaction:

$$\mathcal{L}_{\text{LHEIDIgg}} = \sum_{k=1}^{10} \frac{\sqrt{K_{ggh}} g_s^2}{32\pi^2 v} F(\tau_k) \xi_k H h_k (\partial_\nu G_\mu^a - \partial_\mu G_\nu^a) (\partial^\nu G^{a\mu} - \partial^\mu G^{a\nu})$$

with

$$\tau_k = \frac{4M_T^2}{m_{Hh_k}^2},$$

and the loop form factor as encoded in the file:

$$F(\tau) = \tau [1 + (1 - \tau) f(\tau)],$$

where

$$f(\tau) = \begin{cases} \arcsin^2(\sqrt{1/\tau}), & \text{if } \tau > 1, \\ -\frac{1}{4} \left[\log\left(\frac{1 + \sqrt{1 + \tau}}{1 - \sqrt{1 + \tau}}\right) - i\pi \right]^2, & \text{otherwise.} \end{cases}$$

Here G_μ^a is the gluon field. The **.fr** expression uses only the Abelianized derivative combination $\partial_\nu G_\mu^a - \partial_\mu G_\nu^a$, not the full non-Abelian field strength

$$G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c.$$

LHEIDIggHeavyTop

The heavy-top limit replaces $F(\tau_k) \rightarrow 2/3$:

$$\mathcal{L}_{\text{LHEIDIggHeavyTop}} = \sum_{k=1}^{10} \frac{\sqrt{K_{ggh}} g_s^2}{32\pi^2 v} \frac{2}{3} \xi_k H h_k (\partial_\nu G_\mu^a - \partial_\mu G_\nu^a) (\partial^\nu G^{a\mu} - \partial^\mu G^{a\nu})$$

Equivalently,

$$\mathcal{L}_{\text{LHEIDIggHeavyTop}} = \sum_{k=1}^{10} \frac{\sqrt{K_{ggh}} g_s^2}{48\pi^2 v} \xi_k H h_k G_{\mu\nu,\text{lin}}^a G_{\text{lin}}^{a\mu\nu}$$

with

$$G_{\mu\nu,\text{lin}}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a.$$

LTot

The total Lagrangian exported by the file is

$$\mathcal{L}_{\text{LTot}} = \mathcal{L}_{\text{LHEIDIggHeavyTop}}$$

so the active LTot contains only the heavy-top effective scalar-gluon-gluon interaction, not the full LHEIDI expression.

Field Table

.fr symbol	Class members	Spin	SU(3) rep	SU(2) rep	U(1) charge / hypercharge	Self- conjugate	Mass symbol and value
Hh	Hh1	0	singlet	singlet	0	yes	mhh1 = 120.000000
Hh	Hh2	0	singlet	singlet	0	yes	mhh2 = 141.421356
Hh	Hh3	0	singlet	singlet	0	yes	mhh3 = 223.606798
Hh	Hh4	0	singlet	singlet	0	yes	mhh4 = 316.227766
Hh	Hh5	0	singlet	singlet	0	yes	mhh5 = 412.310563
Hh	Hh6	0	singlet	singlet	0	yes	mhh6 = 509.901951
Hh	Hh7	0	singlet	singlet	0	yes	mhh7 = 608.276253
Hh	Hh8	0	singlet	singlet	0	yes	mhh8 = 707.106781
Hh	Hh9	0	singlet	singlet	0	yes	mhh9 = 806.225775
Hh	Hh10	0	singlet	singlet	0	yes	mhh10 = 905.538514

The values shown are the built-in fallback values from

$$m_{Hh_1} = m_h, \quad m_{Hh_k} = \sqrt{m_b^2 + (k-1)^2 M_c^2} \quad (k = 2, \dots, 10),$$

using

$$m_h = 120, \quad m_b = 100, \quad M_c = 100.$$

Widths are declared as

$$\Gamma_{Hh_k} = \mathbf{whh}k,$$

with fallback numerical value 0 for all modes.

Parameters

Parameter	Default value	Appears in / multiplies	Physical meaning
HeidiV	246.22	Passed to <code>InitHeidi</code> ; TeX v . The explicit Lagrangian terms use the SM symbol v .	Electroweak vacuum expectation value used to initialize the scalar-sector functions.
HeidiMh	120.	Sets the fallback value of <code>mhh1</code> .	Reference Higgs-like scalar mass parameter m_h .
HeidiCs	100.	Enters fallback masses $m_{Hh_k} = \sqrt{m_b^2 + (k-1)^2 M_c^2}$.	Mode-spacing / compactification-scale-like mass parameter M_c .
HeidiMb	100.	Enters fallback masses $m_{Hh_k} = \sqrt{m_b^2 + (k-1)^2 M_c^2}$.	Base scalar mass parameter m_b .
HeidiG2	1.	Passed to <code>InitHeidi</code> ; not explicitly present in <code>LTot</code> .	Scalar-sector coupling-squared input, TeX g_5^2 .
HeidiCutoff	1000.	Declared but not used directly in any displayed Lagrangian expression.	Cutoff parameter Λ .
HeidiNmodes	10	Defines <code>heidiModeRange = Range[10]</code> .	Number of scalar modes retained, $N_H = 10$.
xi[k]	fallback xi[1]=1, xi[2..10]=0	Multiplies each scalar mode in $S = \sum_k \xi_k Hh_k$ and in the effective $Hh_k gg$ interaction.	Mode wavefunction / mixing coefficient controlling the coupling strength of each scalar mode to SM Higgs-like interactions.
kggh	1	Appears as $\sqrt{K_{ggh}}$ multiplying <code>LHEIDIgg</code> and <code>LHEIDIggHeavyTop</code> .	Overall rescaling factor for the effective scalar-gluon-gluon coupling.

Physics Summary

The file encodes ten real gauge-singlet scalar modes `Hh1` through `Hh10` that couple to Standard-Model fields through Higgs-like mixing coefficients ξ_k . The only active total Lagrangian, `LTot`, keeps the heavy-top effective coupling of these scalars to two gluons, so it mediates gluon-fusion production $gg \rightarrow Hh_k$ and the crossed decay $Hh_k \rightarrow gg$.

The last failure the loop worked on

This is the validation report handed to the repair agent at the start of `repair2/round1` — the problem it was asked to fix. The model **passed** after this round; the report below is the state that was repaired, not the final one.

Validation report — model.fr FAILED the tool chain

- FeynRules UFO compile: OK
- Hermiticity check: True
- Kinetic-terms check: True
- Mass-spectrum check: True
- MadGraph import: False
- Heuristic error tags: `selfconjugate_quantum_numbers`, `mg5_import_fail`, `ufo_python_syntax_error`, `mg5_python_traceback`, `mg5_invalid_cmd`

FeynRules check hermiticity output

```
Checking for hermiticity by calculating the Feynman rules contained in L-HC[L].
If the lagrangian is hermitian, then the number of vertices should be zero.
Starting Feynman rule calculation.
Expanding the Lagrangian...
Expanding the indices over 8 cores
Collecting the different structures that enter the vertex.
No vertices found.
0 vertices obtained.
The lagrangian is hermitian.
```

FeynRules check kinetic_terms output

```
Neglecting all terms with more than 2 particles.
All kinetic terms are diagonal.
```

FeynRules check mass_spectrum output

```
Neglecting all terms with more than 2 particles.
All mass terms are diagonal.
Getting mass spectrum.
Checking for less then 0.1% agreement with model file values.
```

FeynRules / Wolfram Engine output (tail)

```
.
From kernel 3 (Local):
Get::noopen: Cannot open Heidi`.
From kernel 4 (Local):
Get::noopen: Cannot open Heidi`.
From kernel 5 (Local):
Get::noopen: Cannot open Heidi`.

General::stop: Further output of Get::noopen will be suppressed during this calculation.
From kernel 1 (Local):
Needs::nocont: Context Heidi` was not created when Needs was evaluated.
From kernel 2 (Local):
Needs::nocont: Context Heidi` was not created when Needs was evaluated.
From kernel 3 (Local):
```

Needs::nocont: Context Heidi` was not created when Needs was evaluated.
 From kernel 5 (Local):
 Needs::nocont: Context Heidi` was not created when Needs was evaluated.
 From kernel 6 (Local):
 Needs::nocont: Context Heidi` was not created when Needs was evaluated.

General::stop: Further output of Context `1` was not created when Needs was evaluated. will be suppressed
 From kernel 2 (Local):
 LoadModel::QN: Warning: Selfconjugated fields should not carry quantumnumbers.
 From kernel 5 (Local):
 LoadModel::QN: Warning: Selfconjugated fields should not carry quantumnumbers.
 From kernel 6 (Local):
 LoadModel::QN: Warning: Selfconjugated fields should not carry quantumnumbers.
 From kernel 8 (Local):
 LoadModel::QN: Warning: Selfconjugated fields should not carry quantumnumbers.
 From kernel 1 (Local):
 LoadModel::QN: Warning: Selfconjugated fields should not carry quantumnumbers.

General::stop: Further output of Warning: Selfconjugated fields should not carry quantumnumbers. will be
 Merging model-files...

Get::noopen: Cannot open Heidi`.

Needs::nocont: Context Heidi` was not created when Needs was evaluated.
 This model implementation was created by
 C. Speckner
 Model Version: 0.1
 For more information, type ModelInformation[].

- Loading particle classes.

LoadModel::QN: Warning: Selfconjugated fields should not carry quantumnumbers.
 - Loading gauge group classes.
 - Loading parameter classes.

Model CHEIDI_gen loaded.

Range::range: Range specification in Range[Heidi\$nmodes] does not have appropriate bounds.

2
 $g_s \text{ HhHeidi\$nmodes} \sqrt{kggh} (\text{del}[G[\mu, a], \nu] - \text{del}[G[\nu, a]$
 Range::range: Range specification in Range[-----
2
48 Pi v

[INFO] Total Lagrangian: LSM + LHEIDIggHeavyTop

2
 $g_s \text{ HhHeidi\$nmodes} \sqrt{kggh} (\text{del}[G[\mu, a], \nu] - \text{del}[G[\nu, a]$
 Range::range: Range specification in Range[-----
2
48 Pi v

General::stop: Further output of Range::range will be suppressed during this calculation.
 [INFO] Running FeynRules consistency checks.
 HEPTAPOD-CHECK-BEGIN: hermiticity

```

Checking for hermiticity by calculating the Feynman rules contained in L-HC[L].
If the lagrangian is hermitian, then the number of vertices should be zero.
Starting Feynman rule calculation.
Expanding the Lagrangian...
Expanding the indices over 8 cores
Collecting the different structures that enter the vertex.
No vertices found.
0 vertices obtained.
The lagrangian is hermitian.
HEPTAPOD-CHECK-END: hermiticity
HEPTAPOD-CHECK-BEGIN: kinetic_terms
Neglecting all terms with more than 2 particles.
All kinetic terms are diagonal.
HEPTAPOD-CHECK-END: kinetic_terms
HEPTAPOD-CHECK-BEGIN: mass_spectrum
Neglecting all terms with more than 2 particles.
All mass terms are diagonal.
Getting mass spectrum.
Checking for less then 0.1% agreement with model file values.
HEPTAPOD-CHECK-END: mass_spectrum
[INFO] UFO output: /tmp/repair_bench2/CHEIDI/round0/UFO (AddDecays -> False)
--- Universal FeynRules Output (UFO) v 1.1 ---
Starting Feynman rule calculation.
Expanding the Lagrangian...
Expanding the indices over 8 cores
Collecting the different structures that enter the vertex.
36 possible non-zero vertices have been found -> starting the computation: Dynamic[FR$FeynmanRules] / 36.
31 vertices obtained.
Flavor expansion of the vertices: Dynamic[FR$Count1] / 31
- Saved vertices in InterfaceRun[ 1 ].
Preparing Python output.
- Splitting vertices into building blocks.
Splitting of vertices distributed over 8 kernels.
- Optimizing: Dynamic[PRIVATE`PY$SplitVertexCounter]/75 .
- Writing files.
Done!
[INFO] Done.

```

MadGraph import output (tail)

```

aMC/models/import_ufo.py", line 251, in import_model
    model = import_full_model(model_path, decay, prefix)
File "$MG5_PATH/models/import_ufo.py", line 413, in import_full_model
    ufo_model = ufomodels.load_model(model_path, decay)
File "$MG5_PATH/models/__init__.py", line 101, in load_model
    raise UFOError(str(error))
models.UFOError: invalid syntax (object_library.py, line 268)
Command "import /private/tmp/repair_bench2/CHEIDI/round0/mg5run/mg5_import.txt" interrupted in sub-command
"import model /tmp/repair_bench2/CHEIDI/round0/UFO" with error:
UFOError : invalid syntax (object_library.py, line 268)
Please report this bug on https://bugs.launchpad.net/mg5amcnlo
More information is found in 'MG5_debug'.
Please attach this file to your report.

```


MG5_debug (the real MadGraph error)

dgraph_interface.py", line 5837, in do_import

```
    raise self.InvalidCmd('UFO model not python3 compatible. You can convert it via the command \nconvert
madgraph.InvalidCmd: UFO model not python3 compatible. You can convert it via the command
convert model /tmp/repair_bench2/CHEIDI/round0/UFO
```

You can also type "set auto_convert_model T" to automatically convert all python2 module to be python3 co

During handling of the above exception, another exception occurred:

Traceback (most recent call last):

File "\$MG5_PATH/madgraph/interface/extended_cmd.py", line 1570, in onecmd

```
    return self.onecmd_orig(line, **opt)
    ~~~~~
```

File "\$MG5_PATH/madgraph/interface/extended_cmd.py", line 1519, in onecmd_orig

```
    return func(arg, **opt)
```

File "\$MG5_PATH/madgraph/interface/master_interface.py", line 281, in do_import

```
    self.cmd.do_import(self, *args, **opts)
    ~~~~~
```

File "\$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5893, in do_import

```
    self.import_command_file(args[1])
    ~~~~~
```

File "\$MG5_PATH/madgraph/interface/extended_cmd.py", line 1718, in import_command_file

```
    self.exec_cmd(line, precmd=True)
    ~~~~~
```

File "\$MG5_PATH/madgraph/interface/extended_cmd.py", line 1599, in exec_cmd

```
    stop = Cmd.onecmd_orig(current_interface, line, **opt)
```

File "\$MG5_PATH/madgraph/interface/extended_cmd.py", line 1519, in onecmd_orig

```
    return func(arg, **opt)
```

File "\$MG5_PATH/madgraph/interface/master_interface.py", line 281, in do_import

```
    self.cmd.do_import(self, *args, **opts)
    ~~~~~
```

File "\$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5835, in do_import

```
    raise err
```

File "\$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5819, in do_import

```
    self._curr_model = import_ufo.import_model(args[1], prefix=prefix,
    ~~~~~
```

```
        complex_mass_scheme=self.options['complex_mass_scheme'],
    ~~~~~
```

```
        options=options)
    ~~~~~
```

File "\$MG5_PATH/models/import_ufo.py", line 251, in import_model

```
    model = import_full_model(model_path, decay, prefix)
```

File "\$MG5_PATH/models/import_ufo.py", line 413, in import_full_model

```
    ufo_model = ufomodels.load_model(model_path, decay)
```

File "\$MG5_PATH/models/__init__.py", line 101, in load_model

```
    raise UFOError(str(error))
```

models.UFOError: invalid syntax (object_library.py, line 268)

Fail to write options with error No model currently active, please import a model!

UFO Python syntax error — UFO/parameters.py line 28: invalid syntax

This generated file is not valid Python, which is why MadGraph's import fails. Find the .fr declaration that produced it and fix THAT (the UFO is regenerated each round).

```
26 |             lhacode = [ 1 ] ]
27 |
```

```
28 | Heidi$v = Parameter(name = 'Heidi$v',  
29 |           nature = 'external',  
30 |           type = 'real',
```

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: ☐ approve ☐ approve with corrections ☐ reject
- Notes: